AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM276
Software Development Processes and Methodologies

AOU EGYPT

Final Examination
Spring Semester 2025/2026

Date: 11 /Jun./2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

Part I – Multiple Choice Questions 5 Marks

*Answer **all** the following multiple-choice questions. Each question is worth 1 Mark.*

1. Which safety achievement states that the system should be designed so that some classes of hazard simply cannot arise.

a. Hazard Avoidance	b. Hazard Detection	c. Hazard Removal	d. Damage Limitation

2. Which of the following is a threat type that allow an attacker to make part of the system unavailable.

a. Interception	b. Interruption	c. Modification	d. Fabrication

3. In which of the following interface types, data is passed from one method or procedure to another.

a. Parameter	b. Shared memory	c. Procedural	d. Message Passing

4. Which dependability property is best defined as "The ability of the system to deliver services as specified"?

a. Availability	b. Reliability	c. Safety	d. Security

5. A "Risk" whose consequences have no major effect and no extra cost should be incurred to reduce the hazard probability is categorized as:

a. Intolerable	b. ALARP	c. Acceptable	d. All of the previous

Part II – Short Answer Questions 25 Marks

*Answer **all** the following short answer questions.*

1. Reuse could be used at different levels. Briefly discuss 3 different Reuse levels.

[6 Marks]

2. Answer the following based on Protection Systems.
   a. Define what Protection Systems are.
   b. Briefly explain how Protection Systems work.
   c. Why are Protection Systems built using Diversity and Redundancy?

[8 Marks]

3. Why does the Dependability costs tend to increase exponentially as increasing levels of dependability are required?
   When is it acceptable for software to be released with low level of dependability and when is it not acceptable.

[6 Marks]

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4. Answer the following based on Software Inspection:
    a. How is Software Inspection different that Software Testing?
    b. Briefly discuss 3 advantages of inspection.

[5 Marks]

Part III – Problem Solving Questions — 20 Marks

*Answer **all** the following problem-solving questions. Each question is worth 10 Marks.*

1. Develop a sequence diagram in the Restaurant domain for the process of placing an order with the following requirements:
    a- Create 4 objects to represent the classes UserInterface, Restaurant, Menu, Table.
    b- Create at least 6 messages as you find suitable.
    c- Use the loop fragment to repeat the actions of selecting an item, customizing the ingredients, and adding it to the order.
    d- Make sure to use self-messages at least once

2. a) Briefly explain the "safety" property of a system. [*2 Marks*]
    b) Compare "safety" and "reliability". [*2 Marks*]
    c) Complete the below table by indicating if each system is categorized as "Primary Safety System" or "Secondary Safety System" and briefly state the rationale (reason). [*6 Marks*]

System	Primary/Secondary	Rationale
Insulin pump control		
Mentcare (*Course material case study*)		
Automotive Airbag System		
Railway Signalling		
Emergency Generator		
Tire Pressure Monitoring		

End of questions

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